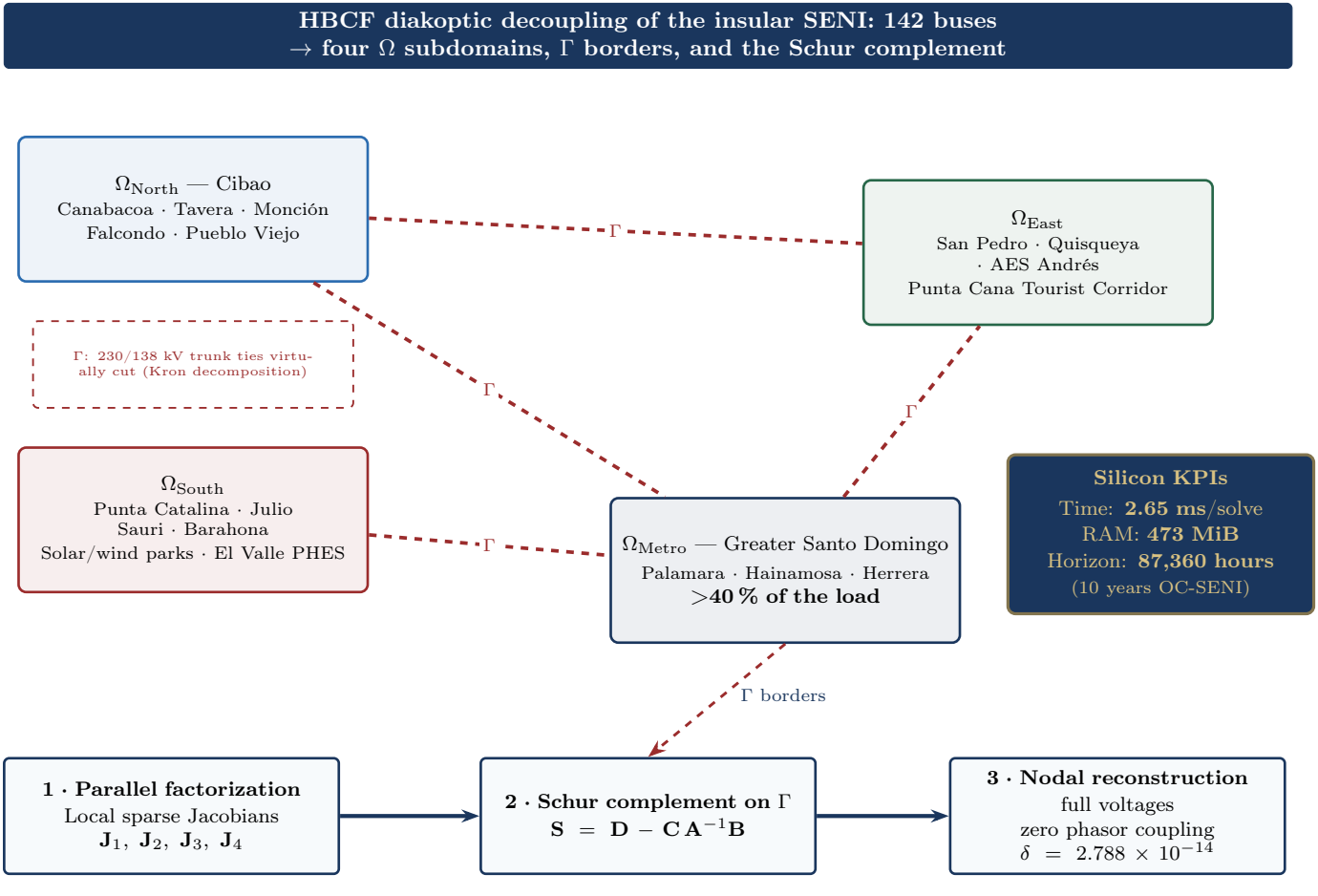

HBCF diakoptic decoupling of the insular SENI: 142 buses
→ four Ω subdomains, Γ borders, and the Schur complement
Ω_{North} — Cibao
Canabacoa · Tavera · Monción
Falcondo · Pueblo Viejo
Ω_{East}
San Pedro · Quisqueya
· AES Andrés
Punta Cana Tourist Corridor
Γ
Γ: 230/138 kV trunk ties virtually cut (Kron decomposition)
Γ
Γ
Ω_{South}
Punta Catalina · Julio
Sauri · Barahona
Solar/wind parks · El Valle PHES
Γ
Ω_{Metro} — Greater Santo Domingo
Palamara · Hainamosa · Herrera
>40 % of the load
Silicon KPIs
Time: 2.65 ms/solve
RAM: 473 MiB
Horizon: 87,360 hours
(10 years OC-SENI)
Γ borders
1 · Parallel factorization
Local sparse Jacobians
$\mathbf{J}_1,\ \mathbf{J}_2,\ \mathbf{J}_3,\ \mathbf{J}_4$
2 · Schur complement on Γ
$\mathbf{S} = \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$
3 · Nodal reconstruction
full voltages
zero phasor coupling
$\delta = 2.788 \times 10^{-14}$